

Review of Revenue Forecasting Model

1. Introduction

2. Objectives

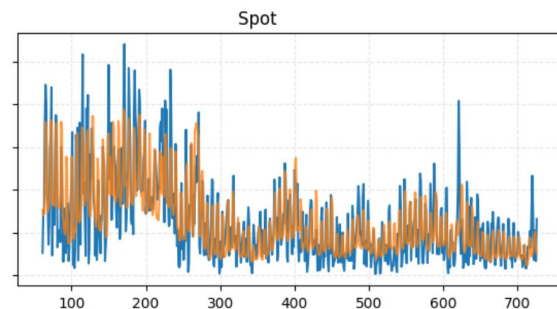
3. Framework	3.1. Methodology	3.2. Data Collection	3.3. Data Preprocessing
	3.4. Feature Selection	3.5. Predictive Modeling	

4. Analyze Results	4.1. Descriptive Statistics	4.2. Feature Validation
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5. Optimization	5.1. Allocation Optimization Algorithm Design	
	5.2. Dynamic Pricing & Coupon Marketing Strategy	

Introduction

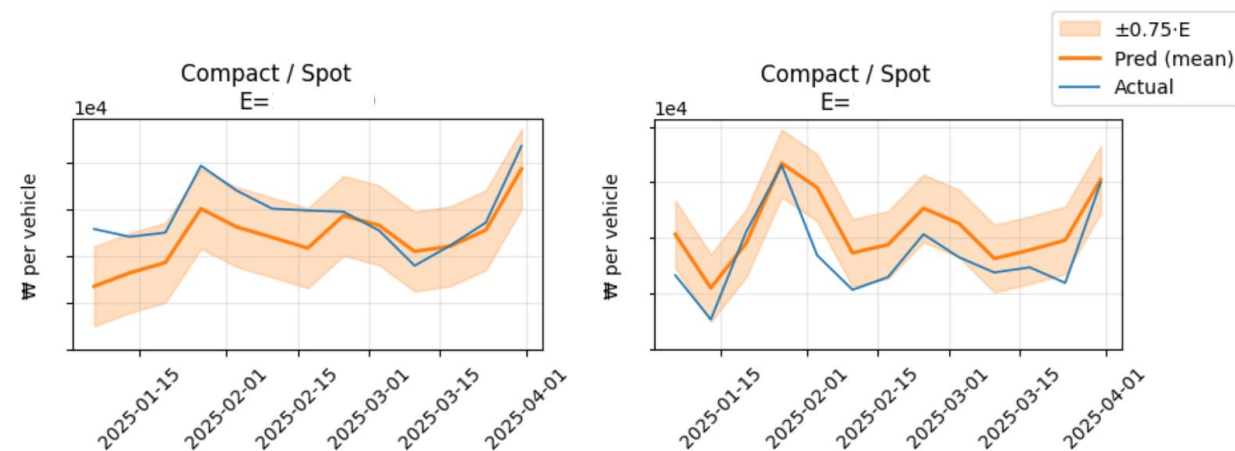
Features
Spot, Vehicle Type, Region Code
Time-Series Data
Weather
Vehicle Information
Utilization Rate
Coupon
Region Information
RPV (Revenue Per Vehicle)



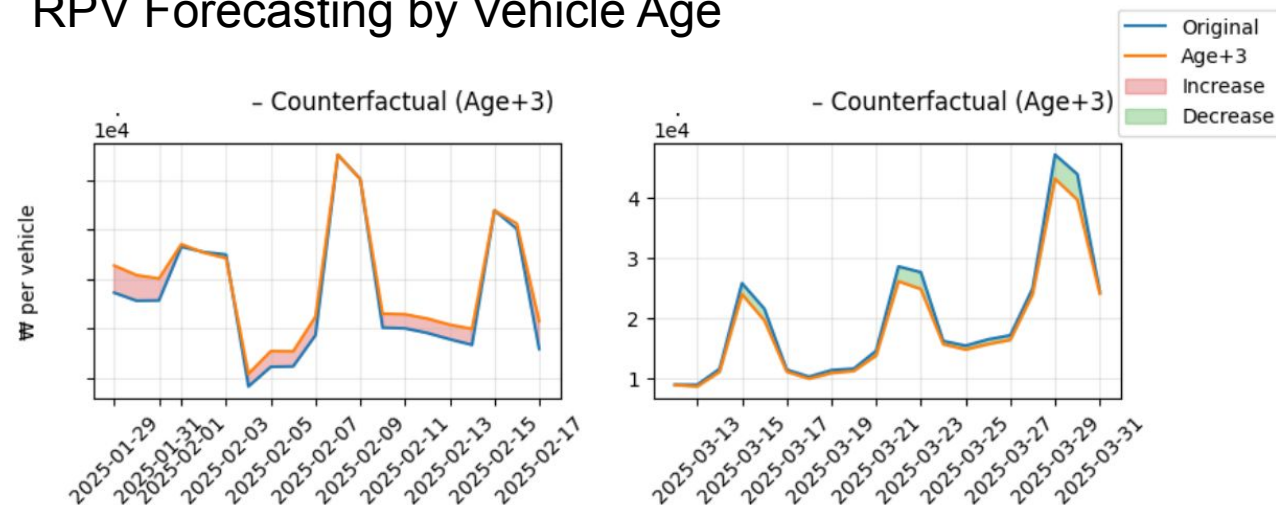
Neural Network

Training on Historical Data

- Weekly RPV Forecasting



- RPV Forecasting by Vehicle Age



Objectives

Designing a Predictive RPV Model for Round-Trips and Exploring Feasibility Based on Transaction Data (2023-2024) and External Factors



Human-Independent, High-Precision Revenue Forecasting Model

Legacy Model

- Historical Average of Similar Profiles
- Human Intuition
- Limited Features
- Validation Limits
- Difficulty in Providing Concrete Metrics

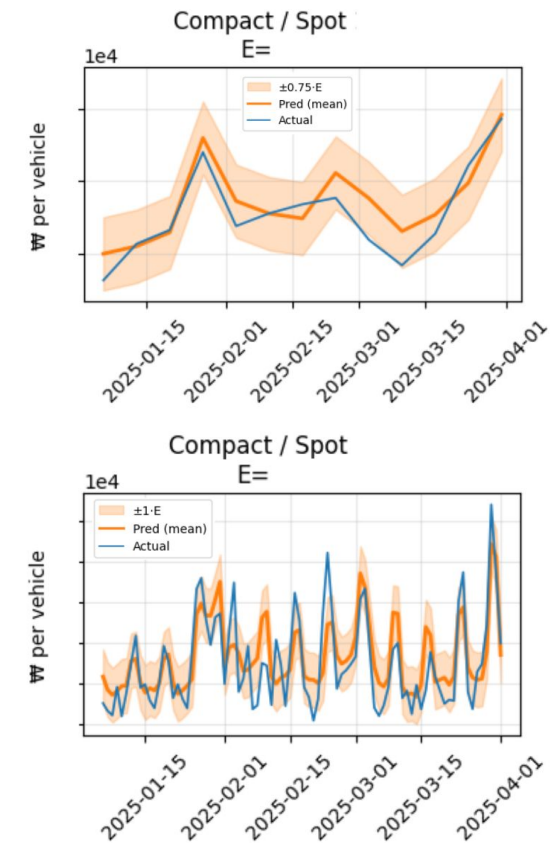


Proposed Model

- Providing Revenue Forecasts & Margin of Error to Partners
- Assessing Revenue Impact via Multi-Variable Scenario Analysis
- Future Data Validation
- Infinite Scalability & Growth Potential

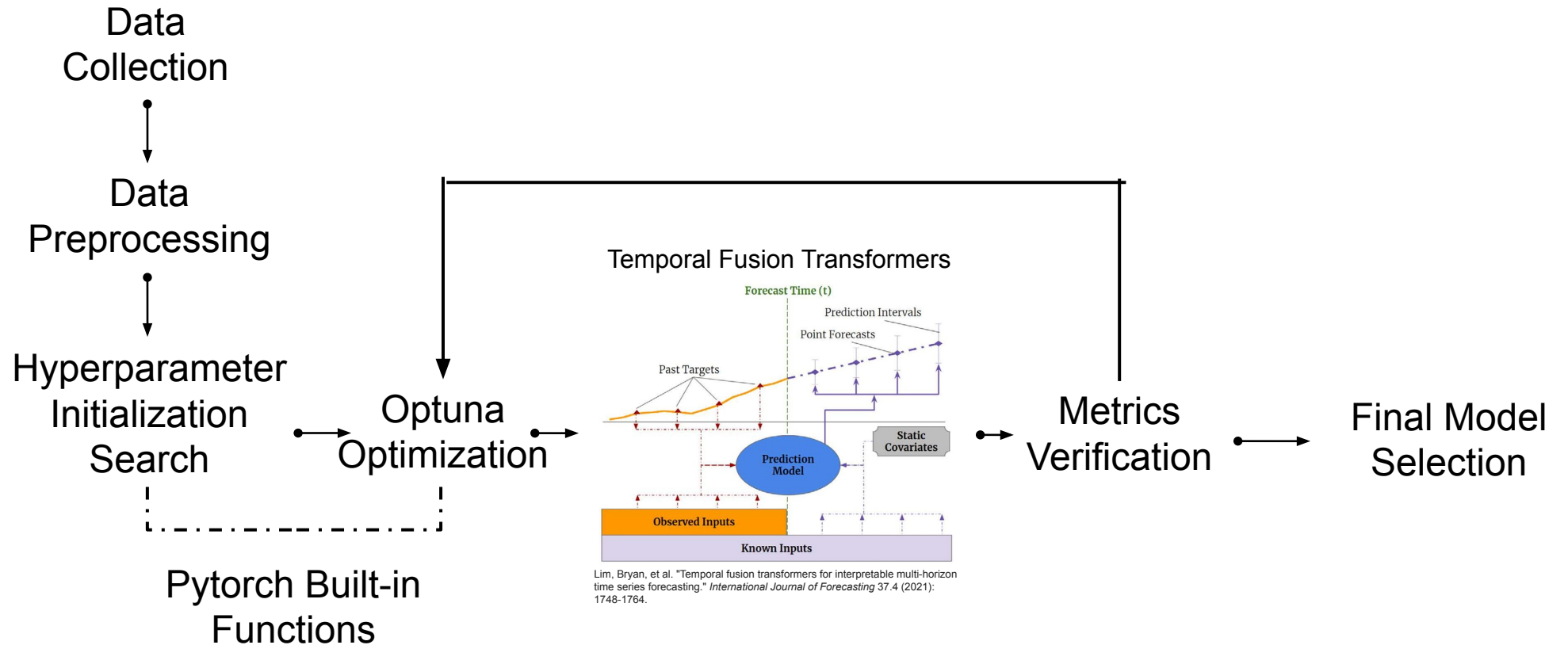


Performance Strategic Reference Level



Framework

(1) Methodology



Framework

(2) Data Collection

2023.01.01 ~ 2025.03.31

Company Internal Data
Revenue, Vehicle, Spot Data

Company Standard Tariff Sheet
by Vehicle Type

KMA Daily Weather API
Temperature, Precipitation, Humidity, Wind
Speed, and Sunshine Duration

MOLIT V-World Spatial Data API
Legal Dong Code & Land Value by Spot

MOIS Population Distribution
by Legal Dong

기상청, 기상(종관, ASOS) 일자료 조회서비스

동관기상관측 장비로 관측한 일 기상자료를 조회하는 서비스

83 3 0

OpenAPI 정보

API 유형: REST, 데이터포맷: JSON+XML, 활용신청: 8728, 등록일: 2019-12-26, 비용부과유무: 무료, 유의사항: 개별단계: 자동승인 / 운영단계: 자동승인

상세기능, 활용사례, 추천데이터

활용명세: Open API 명세 확인 가이드, 가이드 다운로드

행정안전부_법정동별(행정동 통반단위) 성/연령별 주민등록 인구수 1.0.0

[Base URL: apis.data.go.kr/141000/stdgSndgApPnlts]

법정동의 통계년월을 기준으로 시도명, 시군구명, 법정동명, 리명, 행정기관코드, 행정동명, 동, 반, 총인구수, 남자인구수, 여자인구수, 만0~9세남자, ..., 만100세이상남자, 만0~9세여자, ..., 만100세이상여자를 조회하기 위한 서비스입니다.

주민등록 데이터는 통계의 기본이 되는 인구 데이터로써, 주민의 신고에 따라 주민등록법 제6조의 규정에 근거하여 개인·세대별 주민등록표를 기반으로 매월 말일 집계하고 있으며 거주자, 거주불명자, 재외국민을 포함하고 있습니다.

활용가치가 큰 주민등록 데이터의 제공 단위를 통반 단위, 도로명 등으로 확대하여 오픈API 방식으로 제공함으로써, 주민등록 데이터를 활용한 국가정책수립, 민간·공공서비스 개발, 창업 지원 및 일자리 창출 등 다양한 분야의 디지털 전환 및 국민 삶의 질 향상에 기여할 목적으로 작성되었습니다.

API 목록

GET /selectStdgSndgApPnlts 법정부(법정동 통반단위) 성/연령별 주민등록 인구수 목록 조회

표준지공시지가속성조회

국가공인데이터API > 국가공간 개방데이터 > 표준지공시지가정보

표준지의 단위면적당 가격에 대한 속성정보를 조회하는 기능

제공가능 트래픽: 999,999,999

요청주소: https://api.vworld.kr/next/data/getReferLandPriceAttr

제공가능 트래픽: 해당 오픈API를 통해 받은 이용가능한 횟수

요청변수

IdCode: 필수여부: 옵션, 설명: 법정동코드(2~10자리)

stdrYear: 필수여부: 옵션, 설명: 기준연도(YYYY: 4자리)

Company Internal Data System

(3) Data Preprocessing

1. Baseline Data Cleansing

- Daily Revenue Allocation
- Vehicle Inventory Estimation & RPV Calculation by Spot
- Vehicle Spec Mapping via License Plate Number
- Geospatial Mapping via Spot ID

2. Outlier Removal

- Conditional Vehicle Filtering (Total Billing = 0, Insurance Fee = 0)
- Groups with Missing Value Ratio > 60%
- Cases with Zero Revenue

3. Standardization

- GroupNormalizer Scaling for RPV

4. One-Hot Encoding

- Spot, Vehicle Type, Region Code, Time Variable, Holiday Indicator

5. Data Volume

Collected Data	After Preprocessing	Train set	Test set
A	A/4	A/5	A/2000

(4) Feature Selection

Categories	Features
Static Categorical Features	Spot, Vehicle Type, Region Code
Time-Based Deterministic Categorical Features	Day of the Week, Week, Month, Quarter, Weekend & Holiday Flag
Time-Based Deterministic Continuous Features	Proximity to Holiday, Holiday Consecutive Index
	Precipitation, Relative Humidity, Wind Speed
	Vehicle Inventory, Vehicle Availability Status, Average Vehicle Age, Average Standard Tariff
Time-Based Stochastic Continuous Features	1&7&28-Day Lagged Revenue, 7-Day Moving Average & Standard Deviation
	7-Day Utilization Rate, Active Vehicle Count
	Coupon Redemption Count, 1&7-Day Lagged Coupon Redemption Count, Discounted Amount Per Vehicle
	Region Revenue, Region Vehicle Count, Region Revenue Per Vehicle, Region Average Revenue by Vehicle Type
Target Variable	Revenue Per Vehicle

- Spot
- Vehicle Type
- Day of the Week
- Quarter
- Weekend
- Holiday
- Weather
- Vehicle Data
- Moving Average
- Moving Standard Deviation
- Utilization Rate
- Vehicle Inventory
- Coupon
- Region Information
- RPV

Feature Selection Within Data and Time Constraints

(5) Predictive Modeling

- Temporal Fusion Transformer

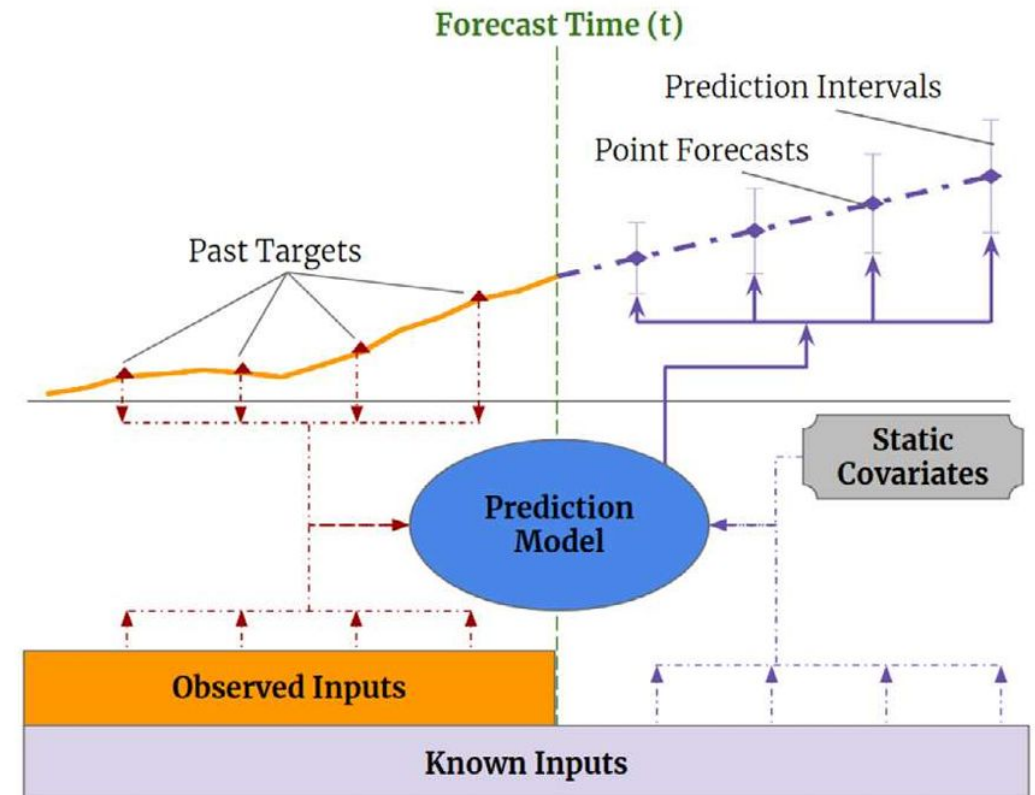
Tri-Categorization of Input Features
(Observed Inputs, Known Inputs, Static Covariates)

Feature Importance Evaluation via Attention Mechanisms

Integration of Exogenous Variables Beyond Traditional
Time-Series

Quantile Regression for Precise Projections and
Confidence Intervals

-> TFT predicts not only **‘when & how much’** but also **‘why’**



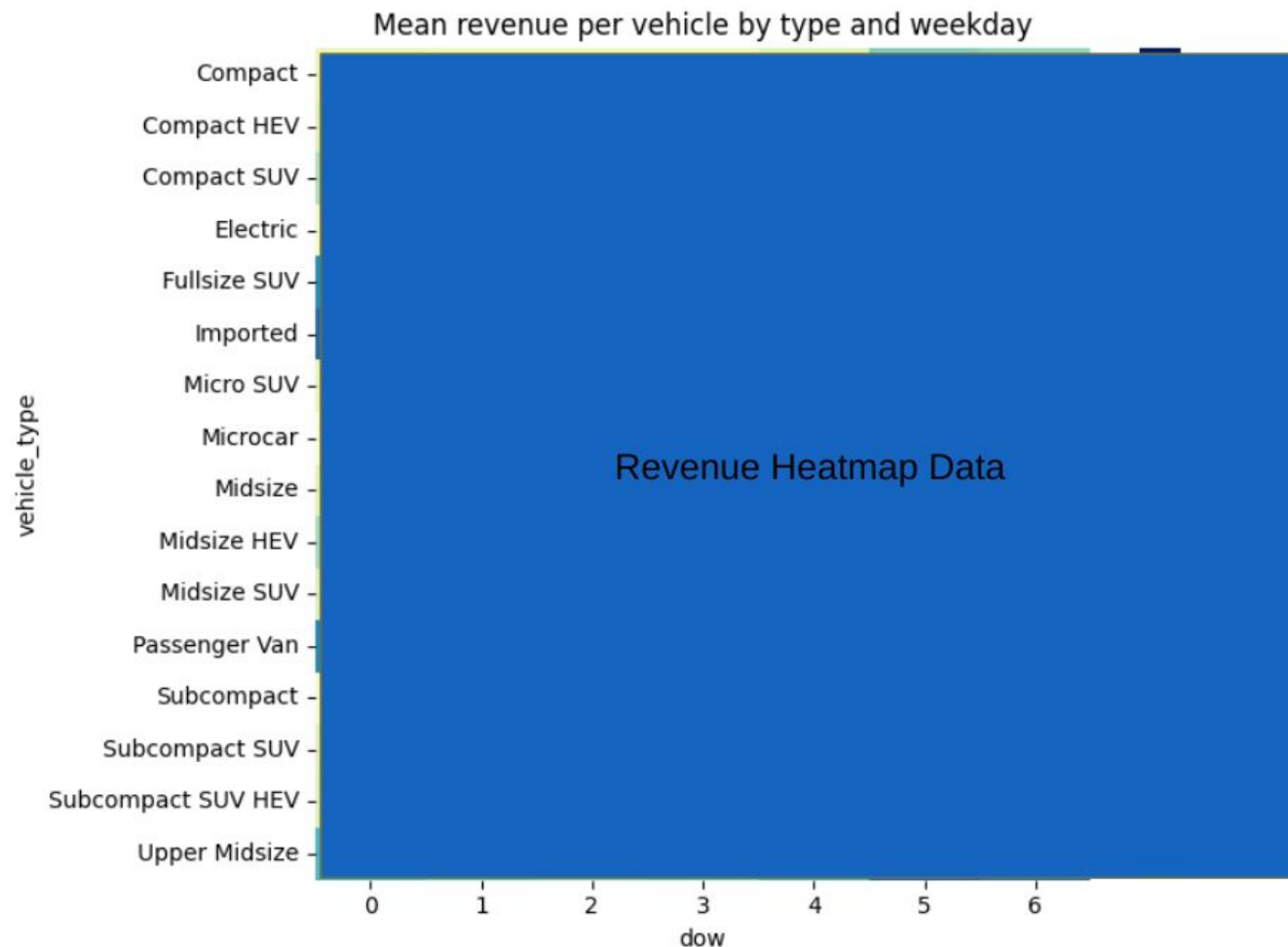
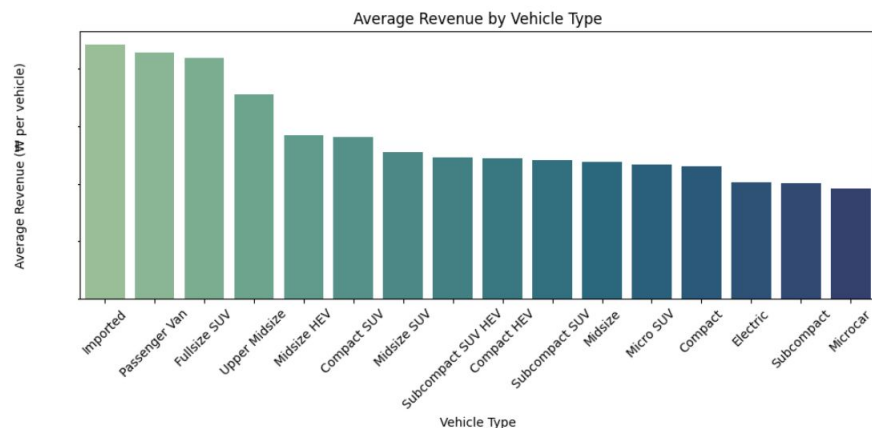
Lim, Bryan, et al. "Temporal fusion transformers for interpretable multi-horizon time series forecasting." *International Journal of Forecasting* 37.4 (2021): 1748-1764.

Analyze Results

(1) Descriptive Statistics – Revenue Trends by Vehicle Type

Summary:

1. Compact vehicles show relatively lower sensitivity to weekend demand spikes.
2. Imported vehicles appear to be largely unaffected by day-of-the-week feature.
3. Weekend generate an significant additional revenue per vehicle compared to weekdays

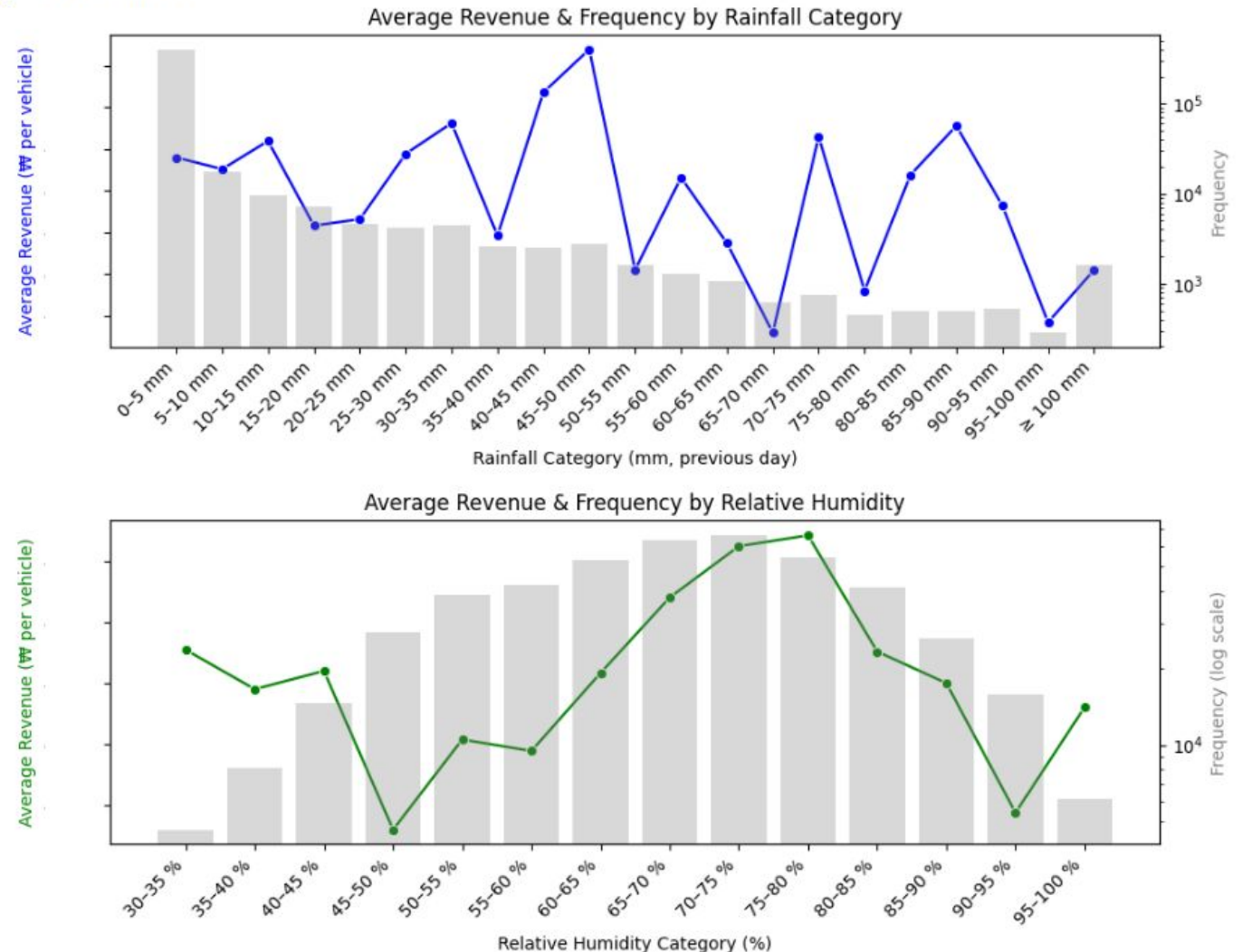


Analyze Results

(1) Descriptive Statistics – Revenue Trends by Weather

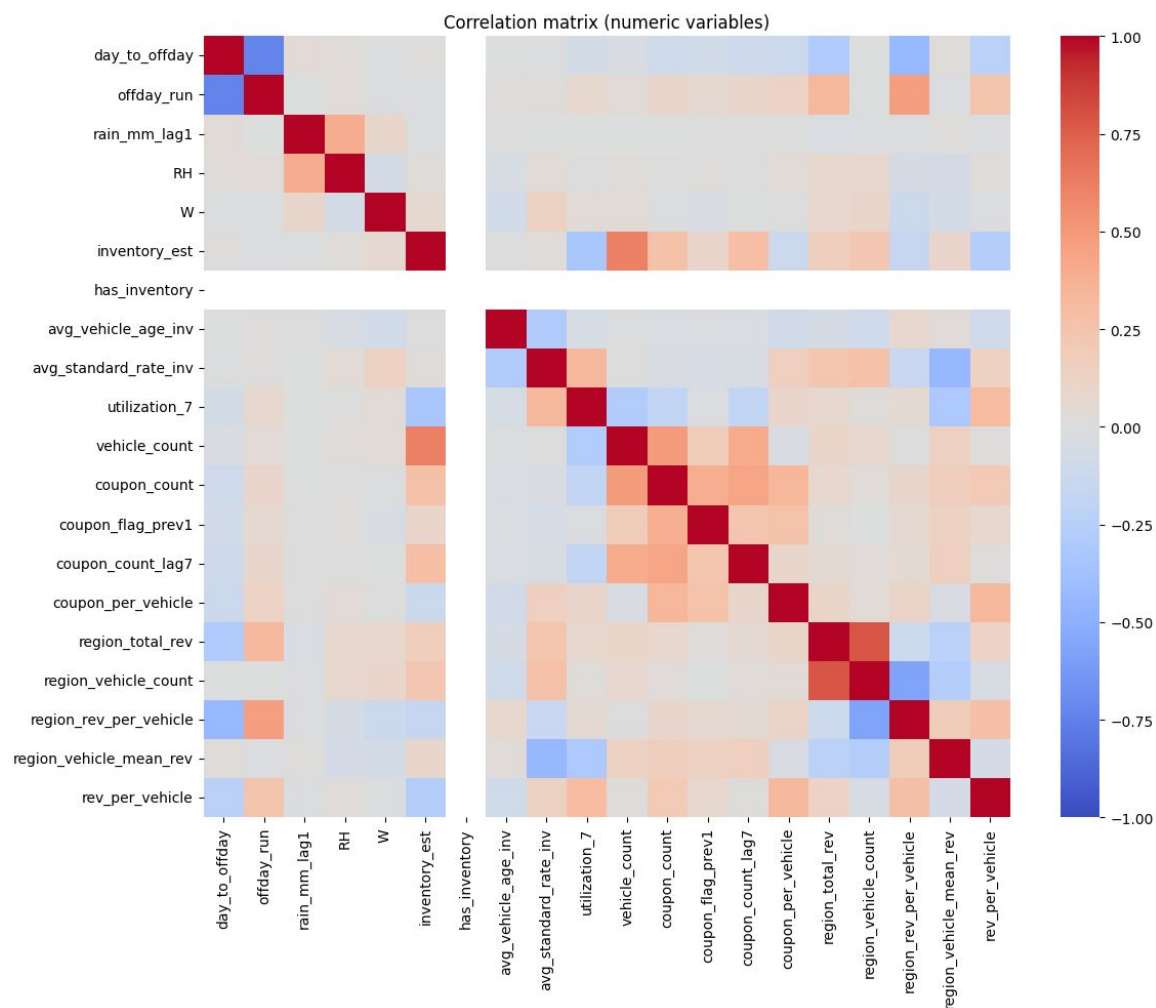
Summary:

1. The direct correlation between same-day precipitation and revenue is ambiguous.
2. However, 1-day lagged precipitation shows a clear negative trend during heavy rain events.
3. While relative humidity strongly correlates with revenue, peak RPV occurs at the 70-80% range rather than the standard comfort zone (40-70%).
4. This suggests a proactive rental pattern where users secure vehicles immediately before and after rain. Also, compounded by peak summer seasonality.



Analyze Results

(2) Feature Validation – Correlation Analyze



Summary:

Correlation Analysis of Independent Variables

While Neural Networks are robust against multicollinearity, excessive feature redundancy can unnecessarily increase model complexity.

Thus, correlation was evaluated to streamline model interpretability.

Most Features appear mutually independent, presenting minimal redundancy.

The two highly correlated cases are derived from intuitive, self-explanatory business logic.

Due to their high interpretability, they pose no threat to training and were intentionally retained

Analyze Results

(2) Feature Validation – LightGBM

Feature Set	Feature Count	WAPE	Contribution
Weather Data	6	0.992391	0.007609
Trend Data	5	0.743138	0.249253
Vehicle Inventory	5	0.737783	0.005355
Coupon	5	0.345268	0.392515
Weather	5	0.345266	0.000002
Land Value, Population	5	0.345287	-0.000021
Region Average	5	0.333713	0.011574
Periodic Features	4	0.330406	0.003307

Summary:

Conducting Forward Feature Ablation to Track Iterative WAPE Reduction Rates

Weather, land price, and population density yield the lowest feature importance, with land price and population showing negative contributions to model accuracy.

While strict computational efficiency dictates the elimination of all three features, weather was retained due to its contextual significance, leading to the strategic exclusion of only land price and population to optimize WAPE.

Analyze Results

(3) Performance Metrics

Quantile Loss

$$L_{\tau}(y, \hat{y}) = \begin{cases} \tau \cdot (y - \hat{y}) & \text{if } y > \hat{y} \\ (1 - \tau) \cdot (\hat{y} - y) & \text{if } y < \hat{y} \end{cases}$$

[0.02, 0.1, 0.25, 0.5, 0.75, 0.9, 0.98]
7-quantile values

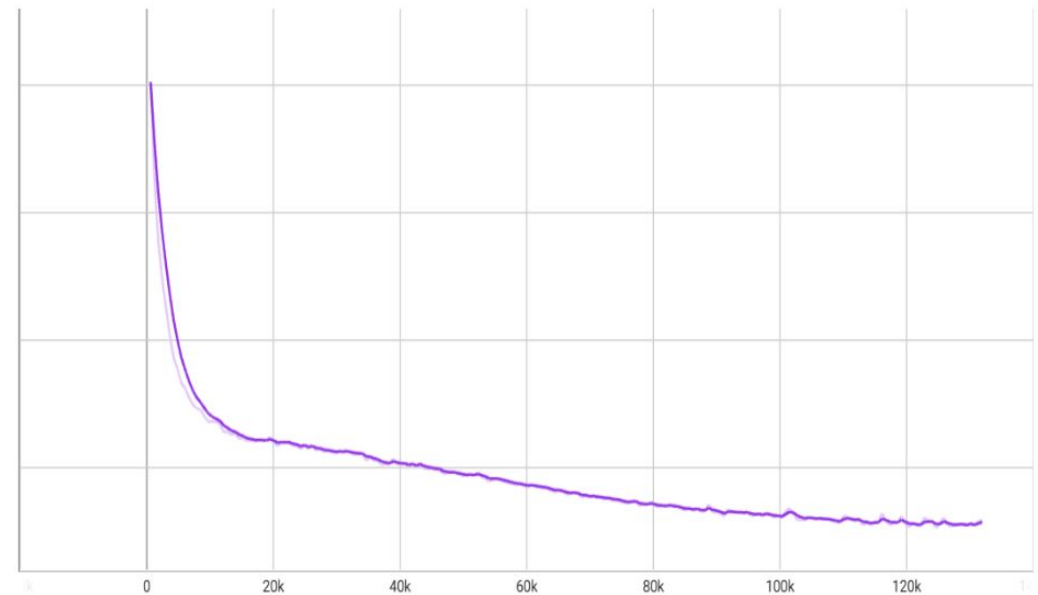
Converging mathematically to 7~98% target quantile values

Optimized by minimizing the joint average of the 7 discrete quantile losses

Predicting not only median (0.5 quantile) but also statistical confidence intervals

-> Differentiating between best-case and worst case demand scenarios to forecast the overall distribution range

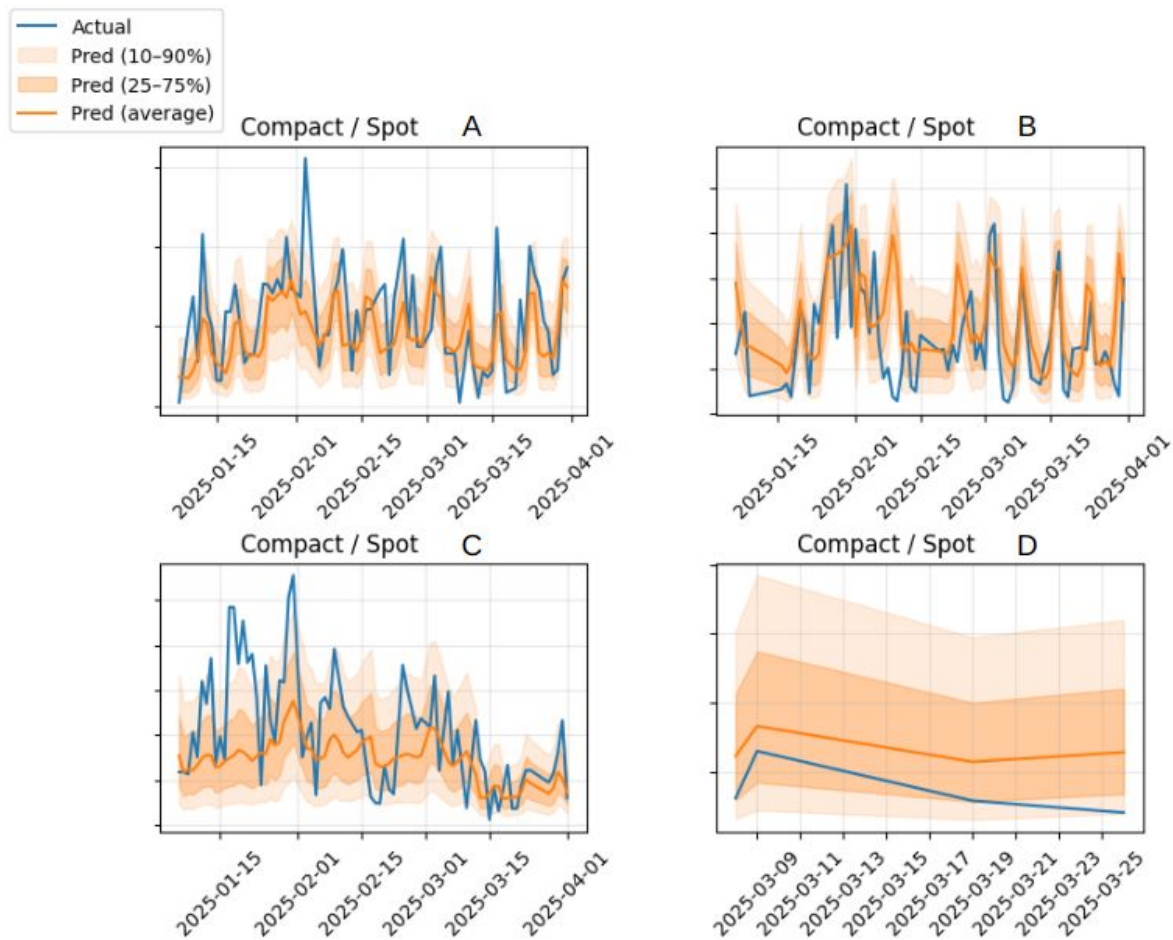
Validation loss



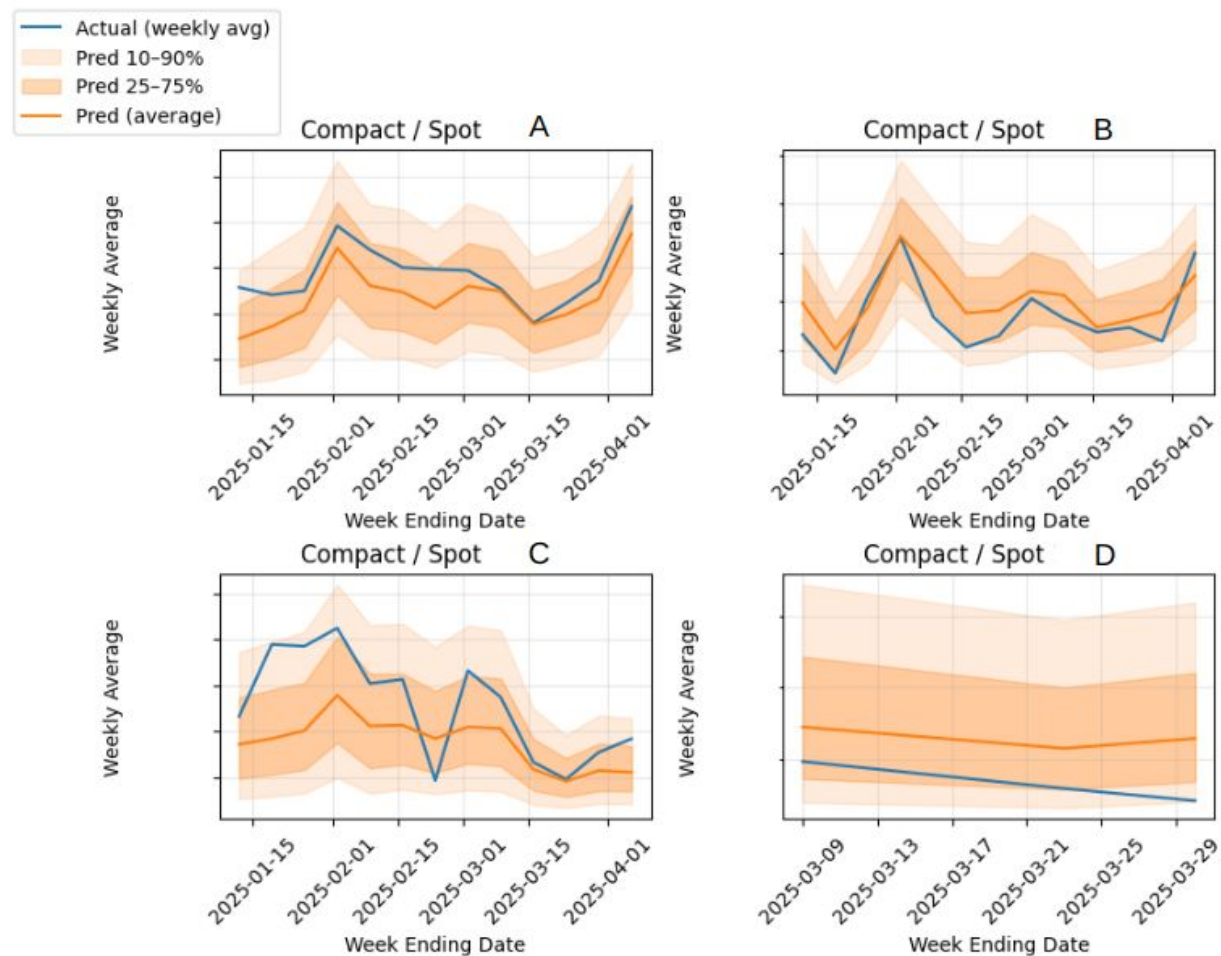
Analyze Results

(4) Use Cases

Compact / Daily Revenue Forecast



Compact / Weekly Revenue Forecast



Analyze Results

(4) Use Cases – RPV Comparison by Vehicle Age

Summary:

Conducting 'What-If' scenario simulations by manually adjusting prospective data to track RPV variance

Revenue Variance when increasing vehicle age by 4 years

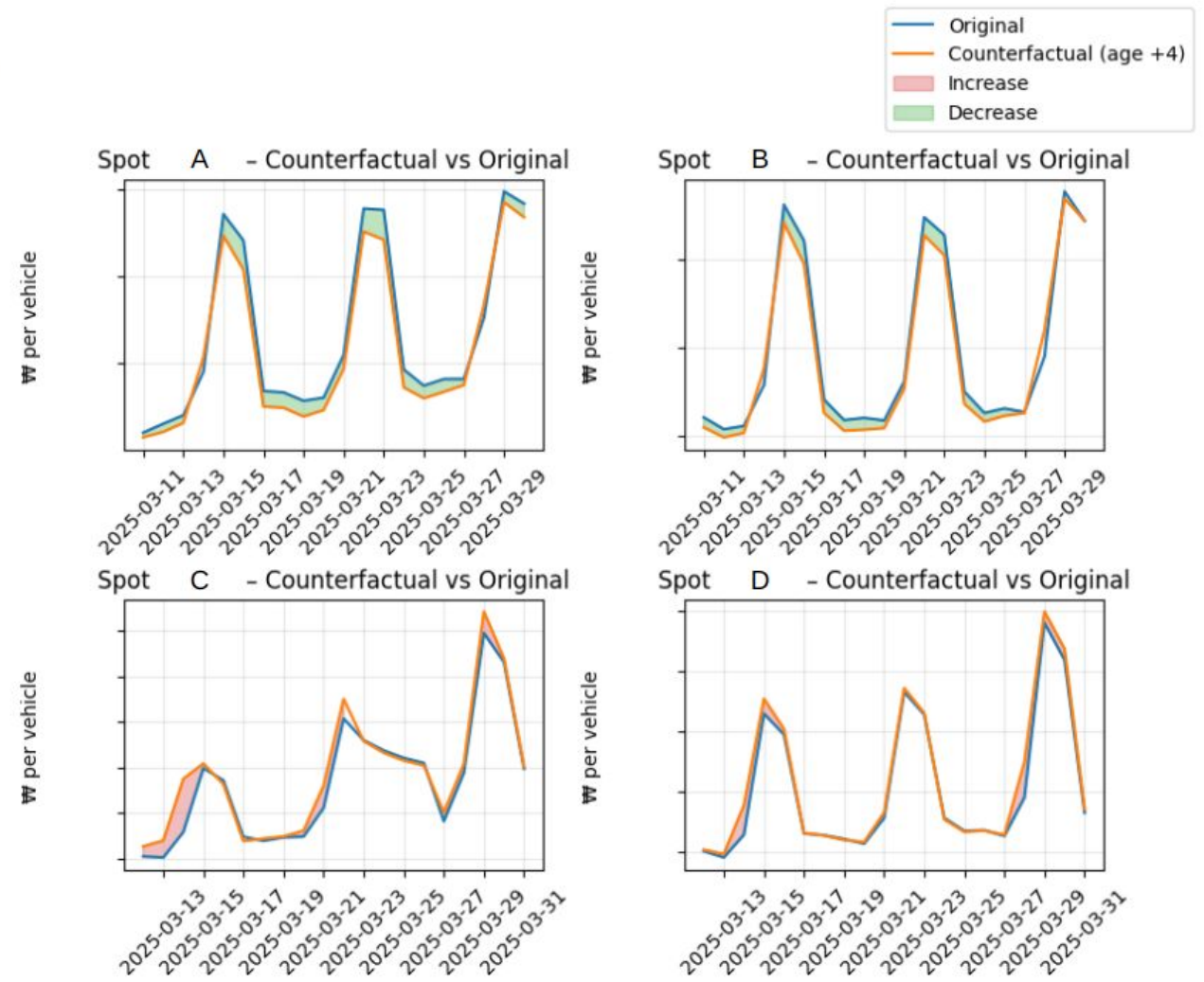
A, B: Residential spot

C, D: High-footfall urban spot

The resulting revenue shift is negligible, remaining within the statistical margin of error

Residential-heavy areas favor newer vehicles, whereas high-footfall zones prefer older vehicles

Core customers in high-footfall zones consist of younger demographics who prefer short-duration rentals and budget friendly pricing.



Analyze Results

(4) Use Cases – RPV Comparison by Vehicle Type

Summary:

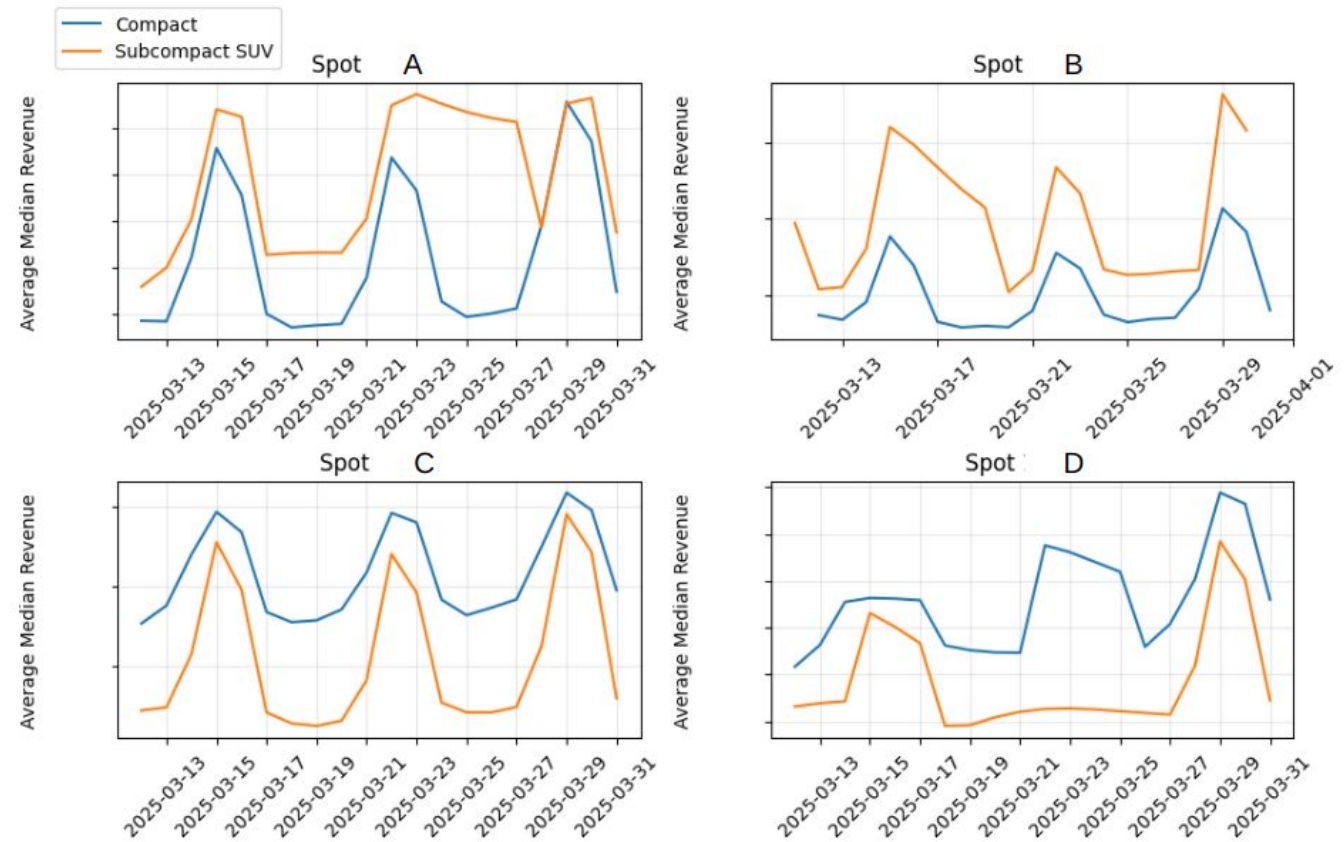
Compact and Subcompact SUV have similar average RPV

A, B: Residential spot

C: University spot

D: Nightlife spot

Small SUVs dominate in family-oriented regions, whereas compact cars outperform in individual centric regions.



Analyze Results

(4) Use Cases – Feature Importance Analytics via Attention Mechanisms

Summary:

From a temporal perspective,
same-day-of-the-week holds critical predictive
value

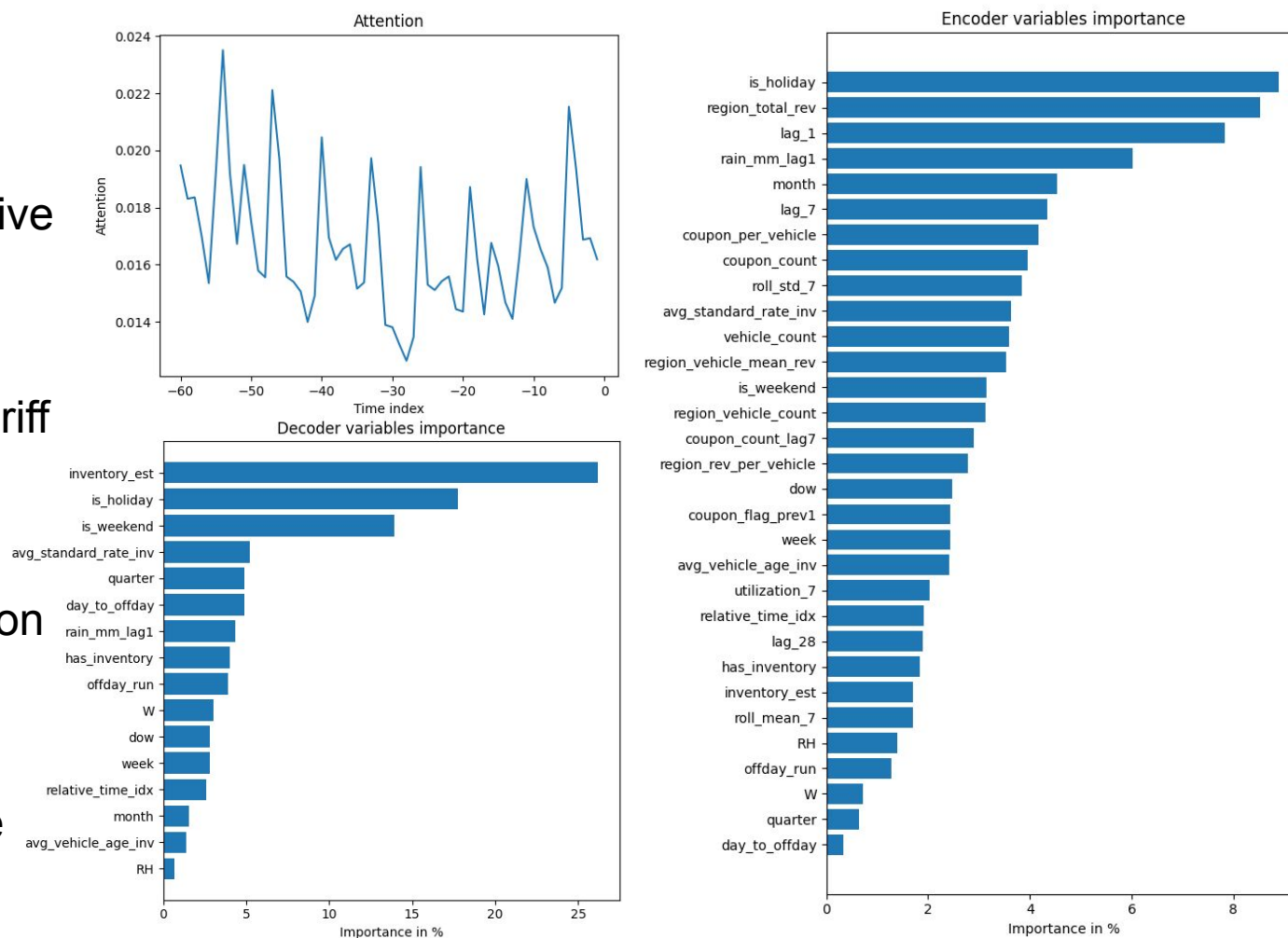
Predictive Priority:

Vehicle Inventory > Holidays > Standard Tariff
> Weather > Temporal Context

Historical Learning Priority:

Holidays > Region Data > Weather > Coupon
Redemptions > Vehicle Count

Contrary to expectations, weather exhibits
marginal impact, while holidays and vehicle
availability dominantly drive revenue.



(1) Available Data Status

Detailed Available Data Information Table
(Including Company Internal Data)

(2) Data Required for Advanced Analytics

Competitors Data

External Paid Data

App Event Data

User geolocation logs,
Spot-specific clickstream data

Abundant
In-App Engagement Indicators Exist



Instead of treating spots as static categorical features, these interest metrics should be vectorized and embedded directly into the modeling pipeline.



Enabling predictive revenue simulation for greenfield spot deployments

Detailed Internal Data

Exact time of car sharing

Precise historical data
of vehicle inventory data
by each spots

Additional Internal Data... etc.

Access to a broader spectrum of internal data will allow the model to ingest diverse features and simulate revenue dynamics across multi-faceted scenarios

Optimization

(1) Allocation Optimization Algorithm Design

Goal: Re-allocate all vehicles in Seoul to maximize total revenue

Simulating dynamic shifts in vehicle availability per station to quantify marginal RPV variance.
Deriving algorithmic vehicle allocation models to optimize geographic supply distribution.
Executing vehicle relocation strictly when the incremental RPV premium exceeds the \$5.0 operational hurdle

Original predicted revenue : A

Predicted revenue after re-allocation : $A \times 115.8\%$

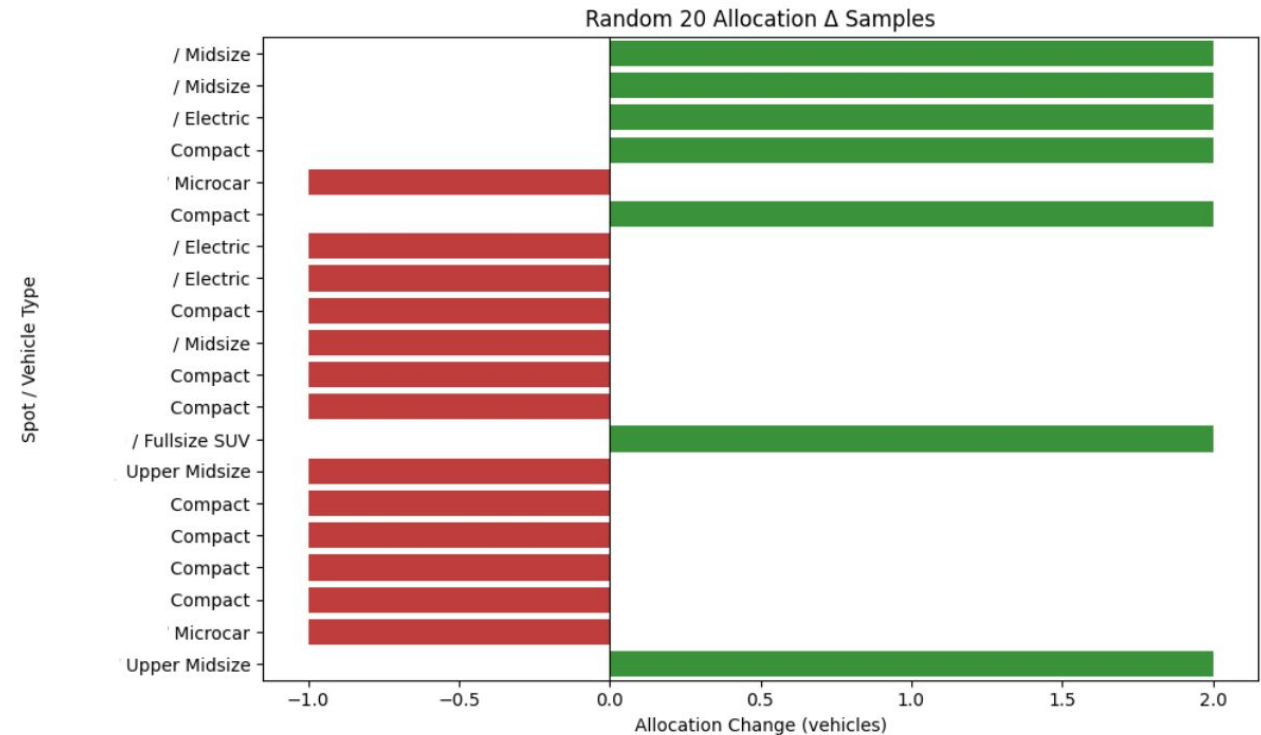
Incorporating spot-specific asset valuations and capital costs into optimization objective.

Precise quantification of inter-spot logistics and vehicle relocation costs.

Maintaining high-fidelity, real-time inventory tracking across all micro-spots.

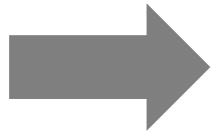
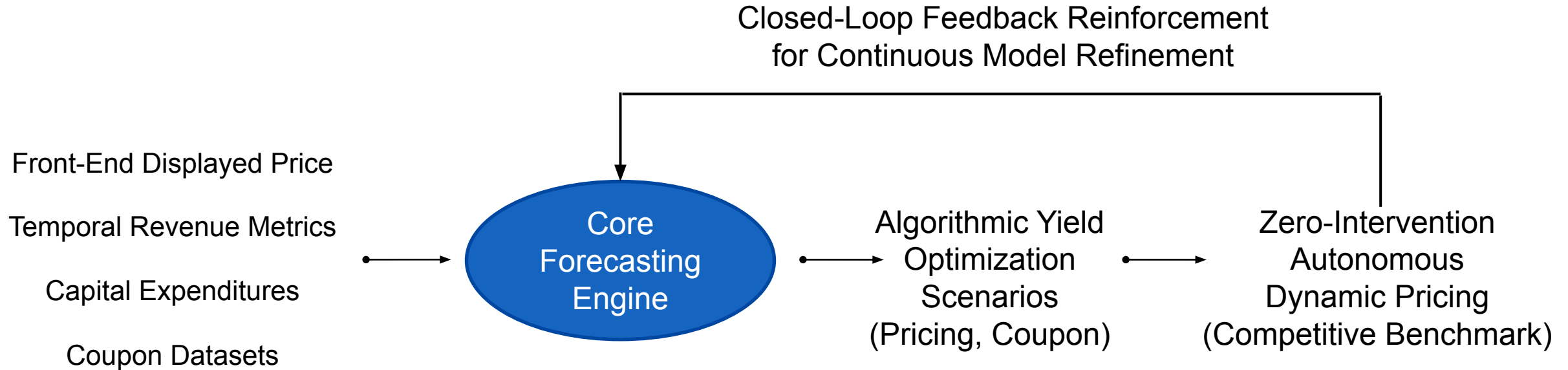


Architecting a high-precision mathematical optimization algorithm for deterministic asset distribution.



Optimization

(2) Dynamic Pricing & Coupon Marketing Strategy



Transcending human intuition to capture real-time market dynamics and maximize net yield
(ex. deploying coupons on the weekends)

Thank you